Propane-Air Westbrook 2-Step Flame Simulation

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This case study illustrates the setup of a propane-air system combustion in a simple coaxial burner. It makes use of the global 2-step mechanism provided by Westbrook et al. (1981). It represents fuel (propane) partial oxidation and the subsequent carbon monoxide/dioxide reactions represented as a pair of irreversible processes. Flow rates are dimensioned to match stoichiometric combustion. Simulations are carried over a 2D axisymmetric domain with OpenFOAM 13 using the `multicomponentFluid` solver with a local time-stepping (LTS) advancement to establish steady-state solutions.

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Introduction

In the present study we simulate propane-air combustion using the 2-step global mechanism proposed by Westbrook (1981). The main goal of this case is providing a robust and fast convergence approach for establishing a steady-state solution in situations where simplified combustion is an acceptable simplification.

Process conditions are provided in Table 1 and burner/domain characteristics in Table 2.

Table 1: Summary of energy input conditions.

Property	Unit	Value
Required power	kW	1000
Lower heating value	MJ/kg	46.3516
Fuel mass flow rate	kg/h	77.6671
Oxidizer mass flow rate	kg/h	1209.86
Total mass flow rate	kg/h	1287.53
Fuel volume flow rate	Nm^3/h	39.4773
Oxidizer volume flow rate	Nm^3/h	939.935
Total volume flow rate	Nm^3/h	979.412
Water production	kg/h	126.918
Carbon dioxide production	kg/h	232.536
Total emissions	kg/h	359.454

Table 2: System dimensions

Quantity	Unit	Value
Fuel inlet diameter	mm	27
Oxidizer inlet inner diameter	mm	32
Oxidizer inlet outer diameter	mm	140
Domain diameter	mm	800
Domain length	m	4

Case is automated to support other operating conditions through `initialConditions` file. Computed quantities related to boundary conditions for fuel and oxidizer are provided in Table 3 and Table 4, respectively. Estimations of turbulent conditions assumed a turbulent intensity of 5%, while characteristic turbulent length scale is taken to be 7% of inlet hydraulic diameter.

Table 3: Conditions evaluated for fuel.

Quantity	Unit	Value
Temperature	K	298.15
Mean velocity	m/s	20.9055
Characteristic length	mm	1.89
Turbulent kinetic energy	m^2/s^2	1.6389
Turbulent dissipation rate	m^2/s^3	182.41

Table 4: Conditions evaluated for oxidizer.

Quantity	Unit	Value
Temperature	K	298.15
Mean velocity	m/s	19.5338
Characteristic length	mm	9.54057
Turbulent kinetic energy	m^2/s^2	1.43089
Turbulent dissipation rate	m^2/s^3	29.4792

An alternative scenario with pre-heated air is also simulated as provided in Table 5. All other conditions for the fuel remain the same.

Table 5: Conditions evaluated for oxidizer.

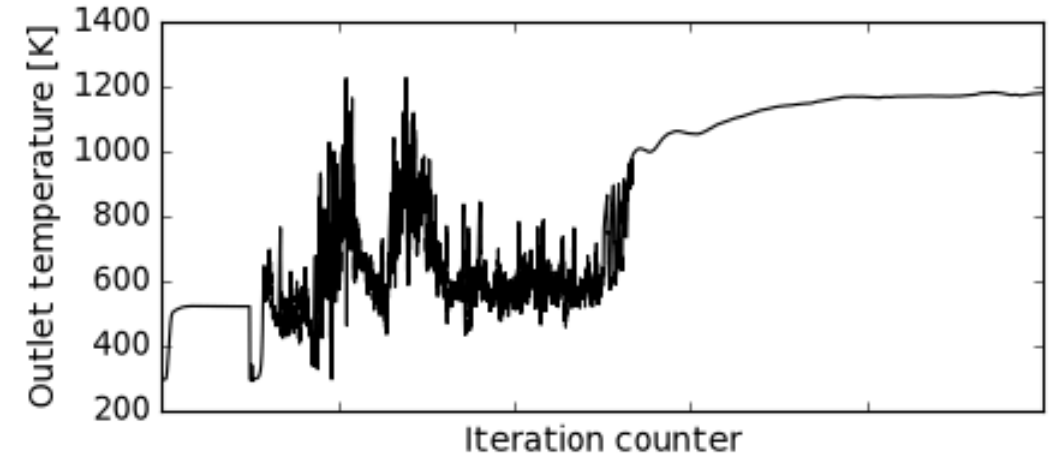
Quantity	Unit	Value
Temperature	K	573.15
Mean velocity	m/s	37.5509
Characteristic length	mm	9.54057
Turbulent kinetic energy	m^2/s^2	5.28776
Turbulent dissipation rate	m^2/s^3	209.419

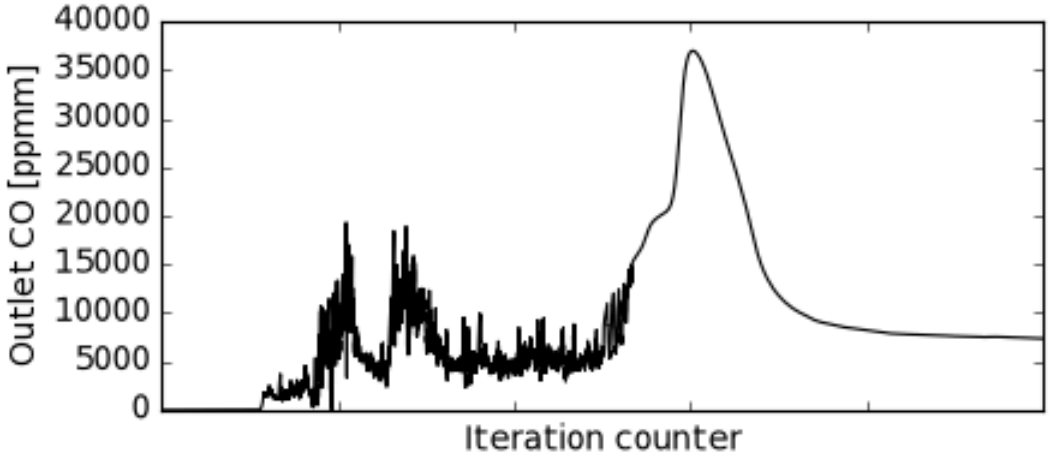
In all cases, walls are modeled through a global heat transfer coefficient representing a refractory of 280 mm (thermal conductivity 1.8 W/(m.K)) sourrounded by steady air at room temperature (heat transfer coefficient 10 $W/(m^2.K)$).

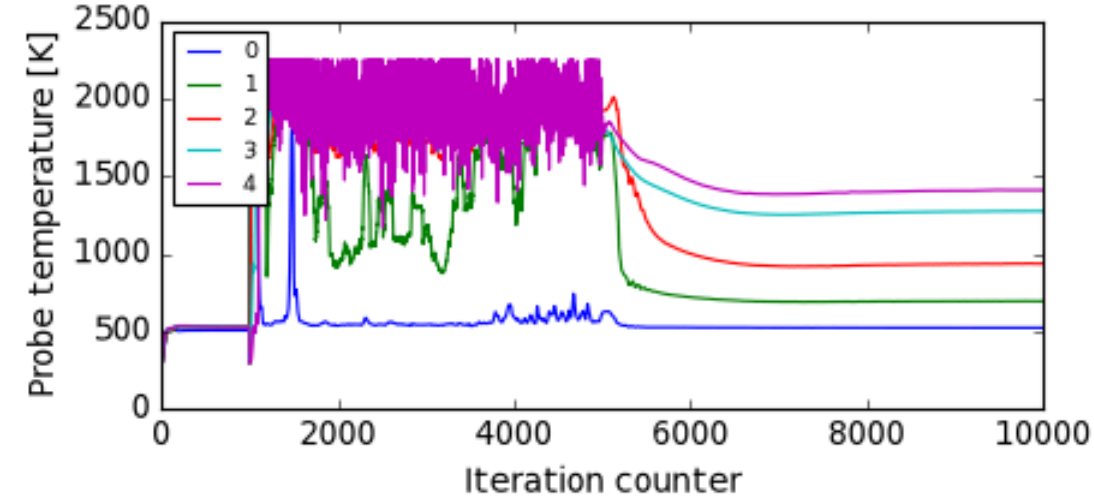
Results

In this section we present a selection of field views for the main variables of interest. All figures are structured the same way, with the reference room temperature results on top, and pre-heated air on the bottom. Care was taken to display both fields with the same color scale.

```
Loading results for iter. 10000.0
Loading results for iter. 10000.0
```


Outlet temperature [K]
1400
1200
1000
800
600
400
200
Iteration counter


Outlet CO [ppmm]
40000
35000
30000
25000
20000
15000
10000
5000
0
Iteration counter


Probe temperature [K]
2500
2000
1500
1000
500
0
0
2000
4000
6000
8000
10000
Iteration counter
0
1
2
3
4

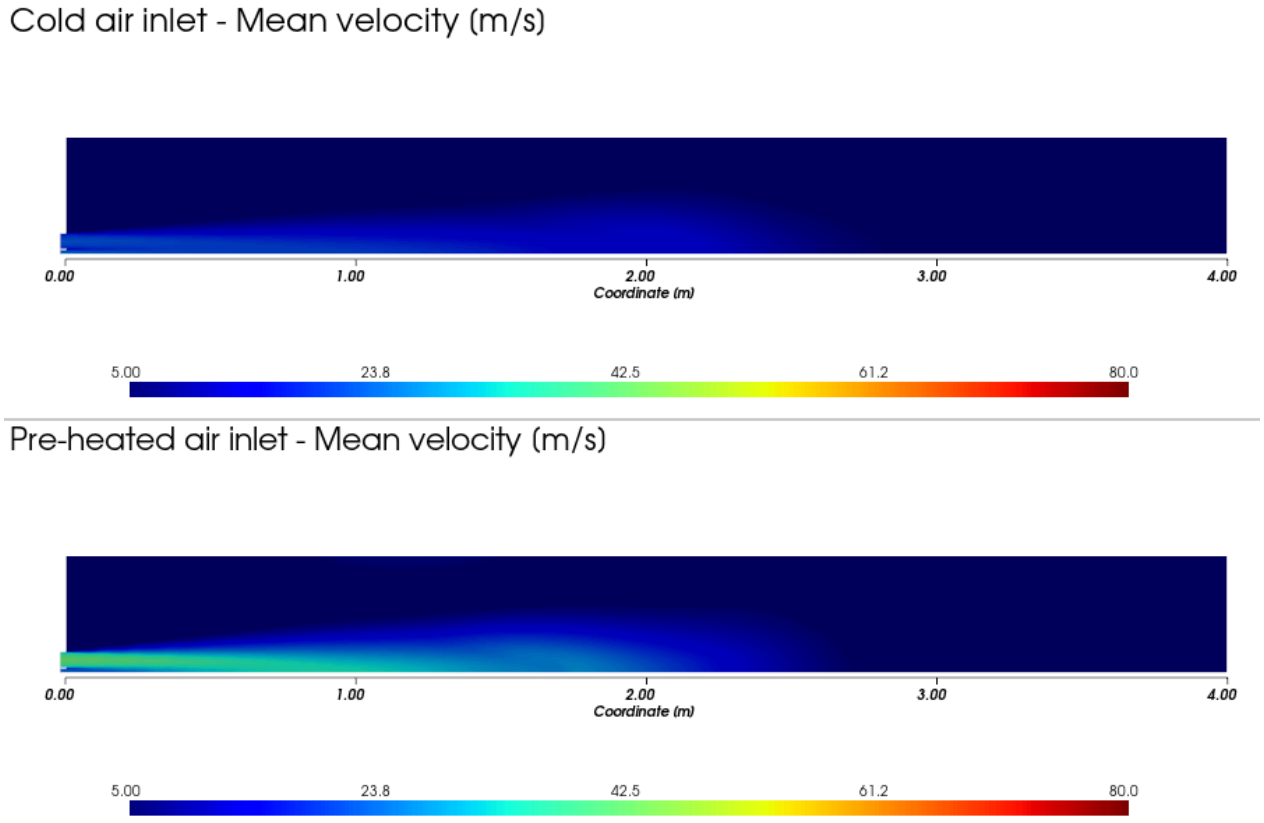


Figure 1: Velocity magnitude fields [m/s].

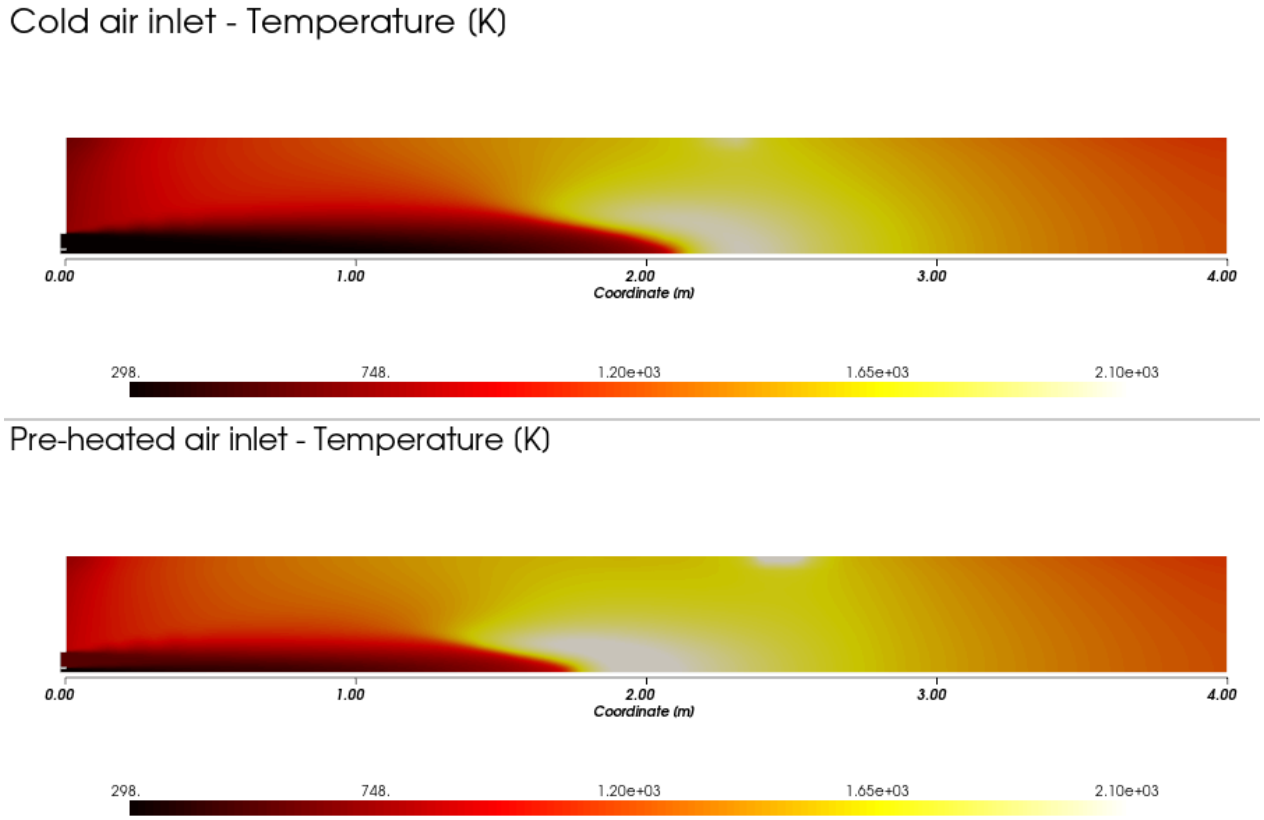


Figure 2: Temperature fields [K].

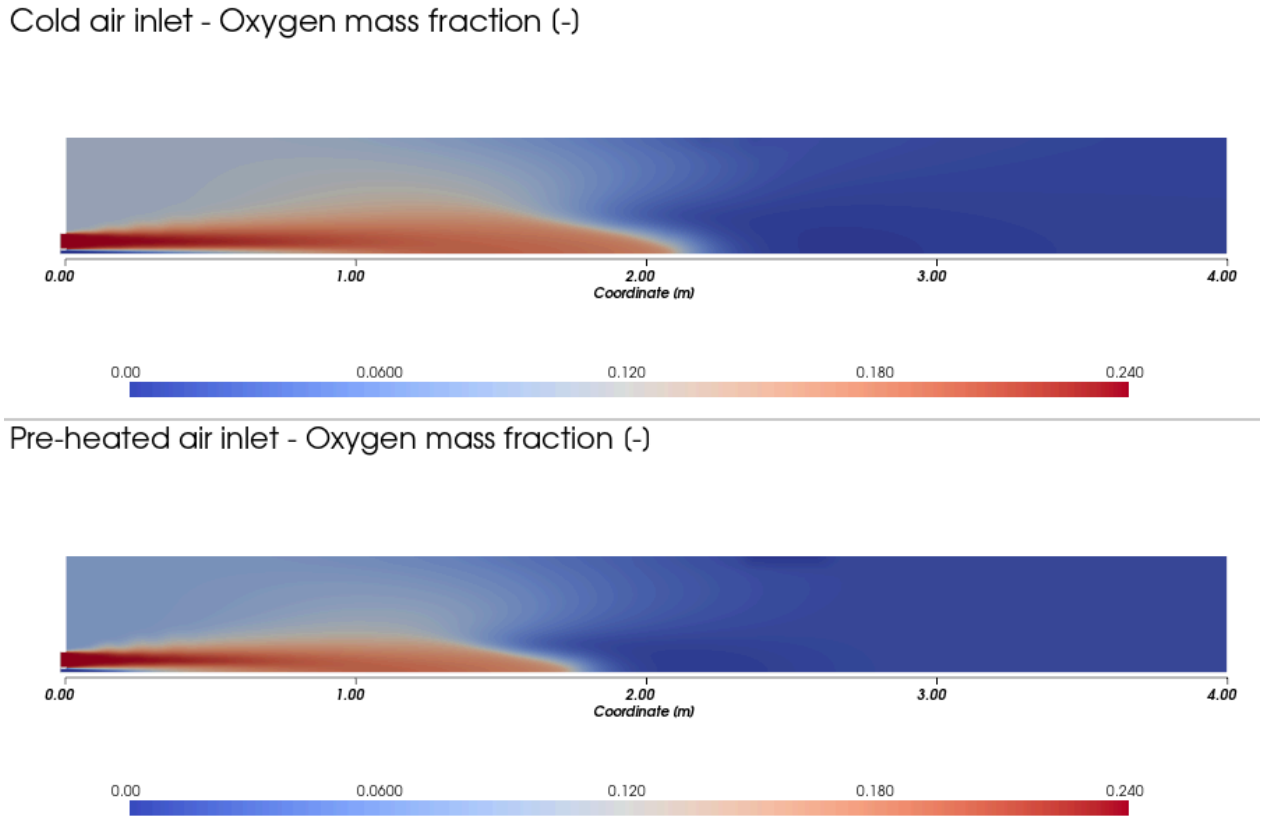


Figure 3: Oxygen mass fraction fields [-].

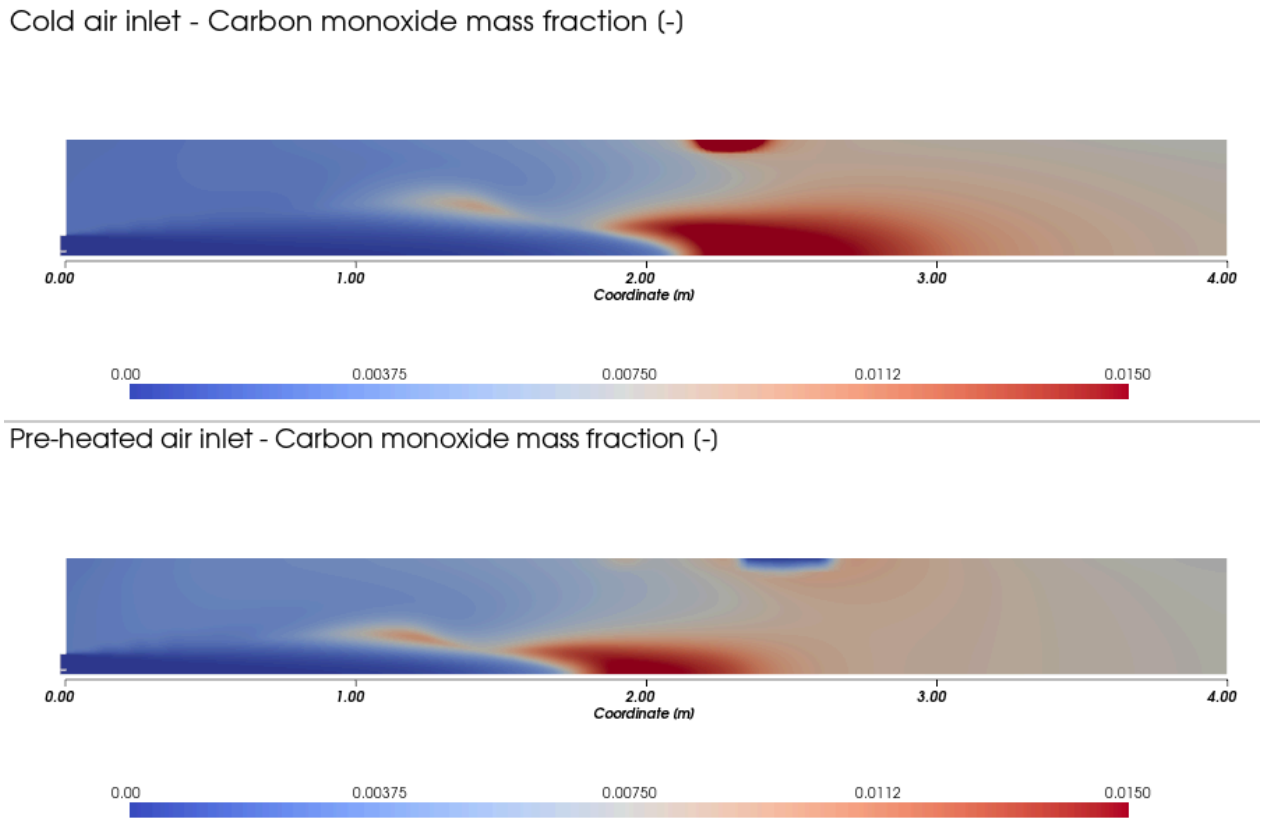


Figure 4: Carbon monoxide mass fraction fields [-].

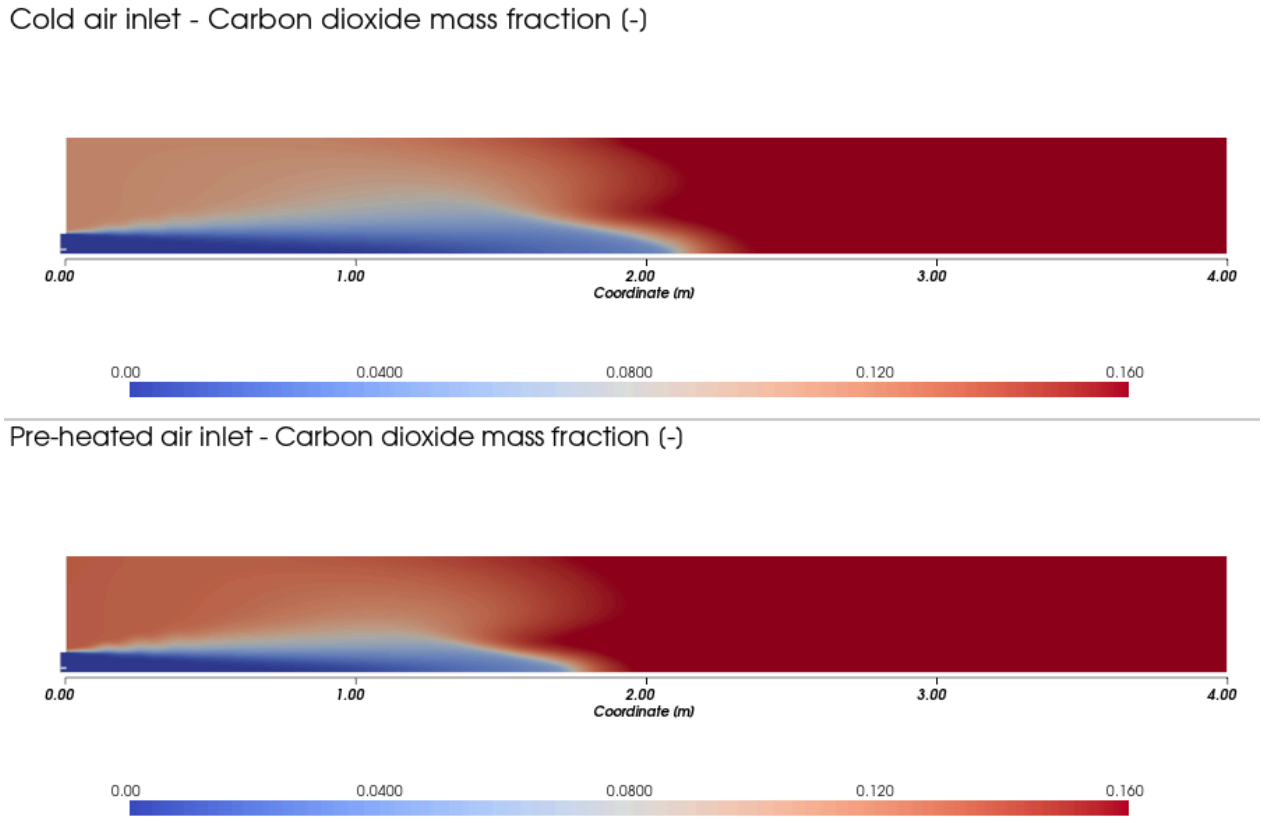


Figure 5: Carbon dioxide mass fraction fields [-].

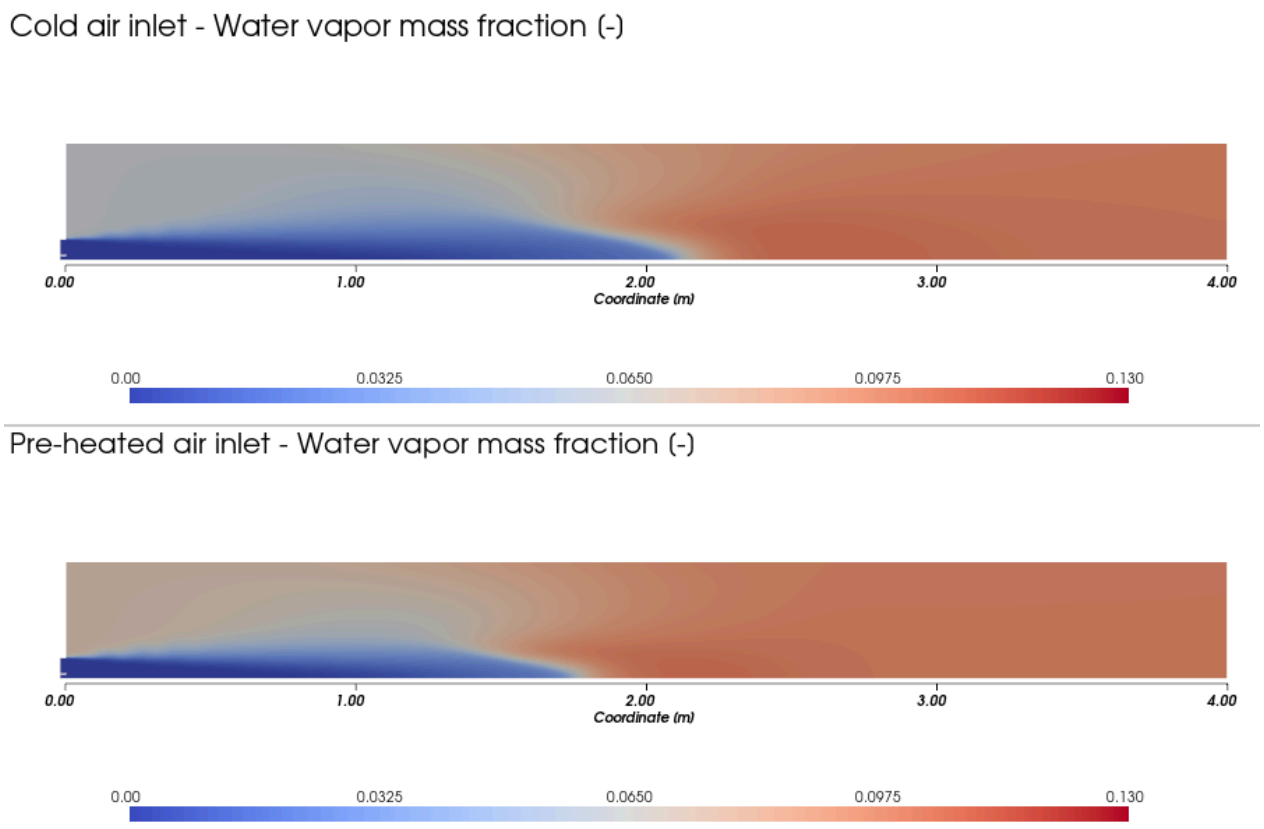


Figure 6: Water vapor mass fraction fields [-].

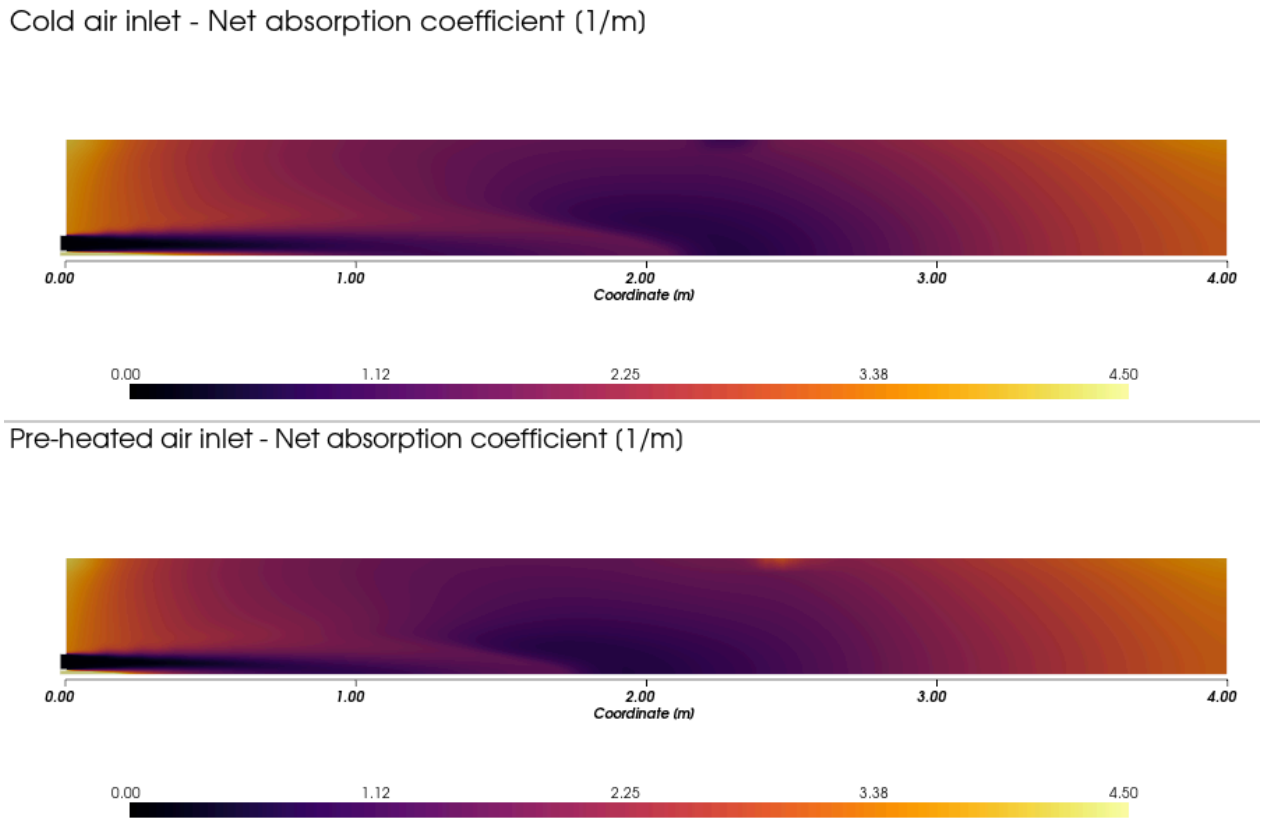


Figure 7: Net absorption coefficient fields [1/m].

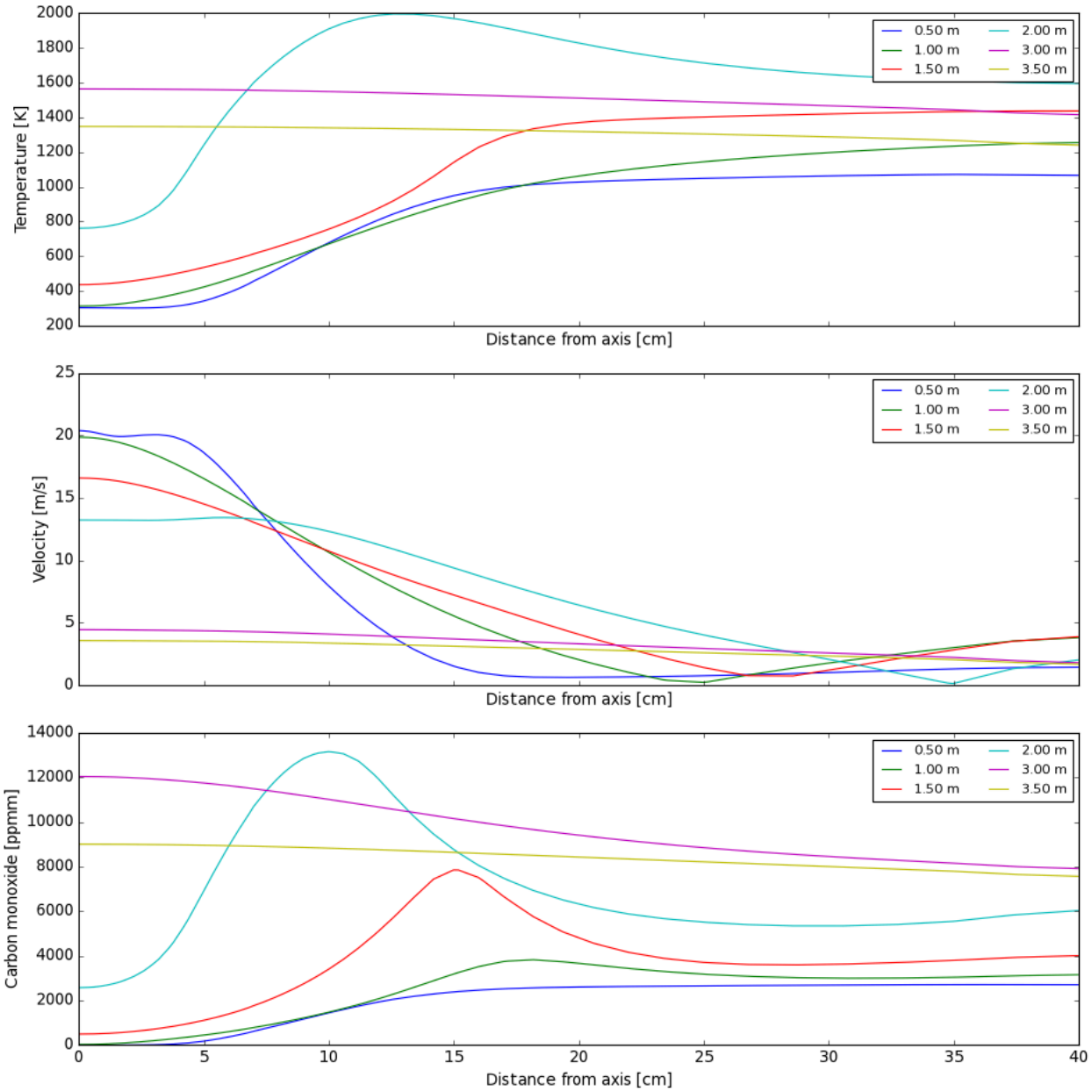


Figure 8: Extraction over selected axial coordinates (room temperature air).

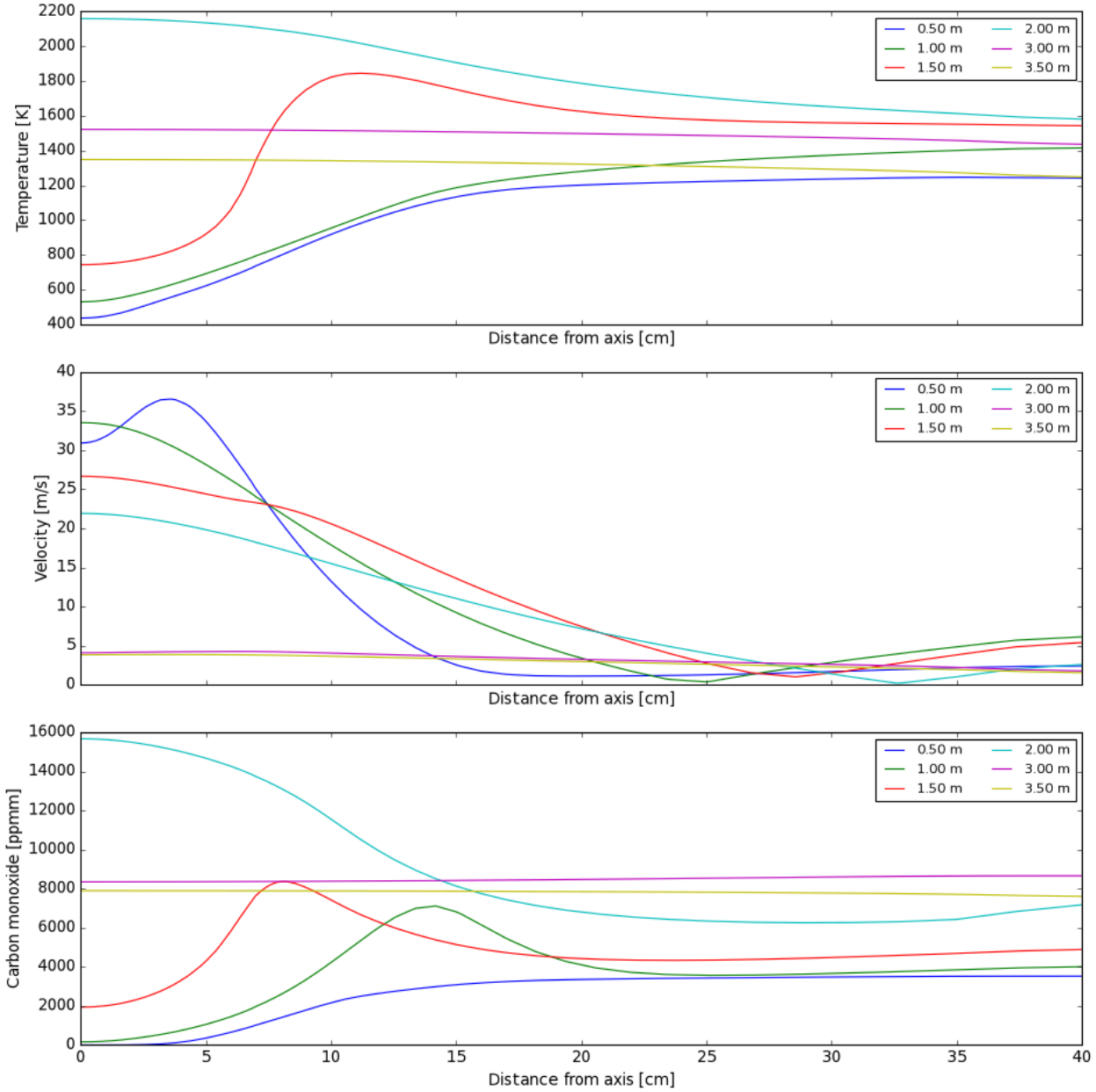


Figure 9: Extraction over selected axial coordinates (pre-heated air).